

Multi-task learning: Train the SLM on multiple related tasks simultaneously (e.g., classification + summarization + question answering). Multi-task learning helps the model generalise better and reduces overfitting on a single task.

Reinforcement Learning with Human Feedback (RLHF): Incorporate human feedback into the training loop to align the model's behaviour with user expectations. For example, OpenAI's fine-tuning of GPT models using RLHF improves reliability and alignment.

Knowledge injection: Inject domain-specific knowledge (e.g., medical guidelines, legal rules) into the SLM to improve reliability in specialised tasks. This can be done in two ways. Use external knowledge graphs to enhance the model's reasoning abilities, and design prompts that guide the model towards reliable outputs.

Strategies for building reliable SLMs

High-quality data curation: Use clean, diverse, and representative datasets for training and fine-tuning. Remove noise and biases during preprocessing.

Domain-specific data: Collect data that reflects the specific domain (e.g., legal, medical, financial). Use synthetic data generation to augment limited datasets.

Robust evaluation metrics: Use task-specific metrics (e.g., F1 score, BLEU score) to evaluate performance. Evaluate robustness using adversarial testing and out-of-distribution data.

Cross-domain testing: Test the SLM on datasets from related domains to assess generalisation.

Explainability tools: Implement explainability techniques to provide insights into model decisions.

- **SHAP (SHapley Additive exPlanations):** Explain feature importance.
- **Attention visualisations:** Show which parts of the input the model focuses on.
- **Counterfactual analysis:** Explores how small input changes affect the output.

Bias detection: Use tools like AIF360 (AI Fairness 360) to detect biases in the training data and model outputs.

Bias mitigation techniques: Reweight training data to balance underrepresented groups. Use adversarial training to reduce bias.

Privacy-preserving techniques: Implement differential privacy to ensure that individual data points cannot be reverse-engineered. Use federated learning to train models on decentralised data without sharing sensitive information.

Security measures: Protect against adversarial attacks by training on adversarial examples. Monitor for data poisoning during training.

Continuous monitoring and updates: Deploy monitoring systems to track the model's performance in real-world scenarios. Continuously update the model with new data to

adapt to changing trends or requirements.

Challenges in ensuring reliable AI with SLMs

Data limitations: Lack of high-quality, domain-specific data can limit the model's reliability. Over-reliance on biased or incomplete datasets can lead to errors.

Overfitting to domain-specific data: Fine-tuning on small datasets can cause the model to overfit and lose generalisation.

Explainability vs complexity: Transformer-based models are inherently complex, making it difficult to explain their decisions.

Scalability: Training large SLMs requires significant computational resources, which may not be feasible for all organisations.

Bias amplification: SLMs can amplify biases present in domain-specific data, leading to unfair or unreliable outcomes.

Creating a specialised language model using multiple LLM sources involves meticulous data collection, preprocessing, filtering, and model training. Microsoft Azure's native services can be leveraged to build a robust and scalable architecture for deploying these models. However, ensuring data quality and reliability is crucial for building an effective SLM. **END** 🐼

References

- https://www.splunk.com/en_us/blog/learn/language-models-slm-vs-llm.html
- <https://azure.microsoft.com/en-us/resources/cloud-computing-dictionary/what-are-small-language-models>
- <https://www.microsoft.com/en-us/microsoft-cloud/blog/2024/11/11/explore-ai-models-key-differences-between-small-language-models-and-large-language-models/>
- <https://www.arionresearch.com/blog/g9xwews0y1yw9ddhkguww06qjf6nz4>
- <https://medium.com/@eastgate/advantages-of-small-language-models-over-large-language-models-a52deb47d50b>
- <https://aisera.com/blog/small-language-models/>
- <https://www.ataccama.com/blog/small-language-models/>
- <https://www.getguru.com/fi/reference/small-language-models>
- <https://blog.nashtechglobal.com/understanding-small-language-models-slms/>

By: Dr Magesh Kasthuri and Dr Anand Nayyar

Dr Magesh Kasthuri is a senior distinguished member of the technical staff and principal consultant at Wipro Ltd. This article expresses his views and not that of Wipro.

Dr Anand Nayyar is a PhD in wireless sensor networks and swarm intelligence. He works at Duy Tan University, Vietnam, and loves to explore open source technologies, IoT, cloud computing, deep learning, and cyber security.